

Modelling of Magnetically Confined Fusion Plasmas: Dimensionless Cross-Field Diffusion in the Souza Tokamak Escape Map

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October 2026

Abstract

This work extends the deterministic tokamak escape map investigated by Souza et al. by introducing a dimensionless model of isotropic cross-field diffusion. The collisionless Souza map and its existing phenomenological stochastic perturbation are first used as reference models. A third model is then constructed in which stochastic motion between deterministic map kicks is represented by Cartesian Brownian increments with dimensionless diffusion strength $\hat{D}$. Deterministic phase evolution and diffusion are combined using symmetric Strang splitting. Repeated Monte Carlo trajectories are used to calculate finite-time Left- and Right-exit probabilities, an exit-conditioned committor, stochastic exit entropy, and a measure of the width of the stochastic ambiguity layer surrounding the deterministic escape-basin boundary.

The implementation reproduces the deterministic map exactly in the zero-diffusion limit and passes independent diffusion, convergence, reproducibility, and probability-bookkeeping tests. At weak diffusion, stochastic uncertainty remains strongly organised by the deterministic basin geometry. An appreciable broadening of the ambiguity layer becomes visible around $\hat{D} \sim 10^{-4}$ within the present dimensionless scan. At larger diffusion strength the Left/Right exit probabilities become increasingly mixed, the exit entropy approaches its maximum over a substantial fraction of the domain, and the predictive influence of the deterministic boundary is reduced. These results demonstrate a crossover from deterministic-like escape to increasingly stochastic transport while retaining a computationally simple reduced-map description.

1 Introduction

The continuous rise in demand for energy has demonstrated the necessity for the development of sustainable sources. The global energy demand increased by an average of 1.3% between 2013-2023, and by 2.2% in 2024.¹ In 2025 the demand lowered to 1.4%, mainly due to the rapid increase in more efficient technologies.² Despite this, the global energy production still increased by 1.7%.² In the same year, 86.2% of energy production came from fossil fuels¹, releasing carbon dioxide and other greenhouse gases into the atmosphere, largely contributing to the deteriorating climate of our planet. Although common renewable sources such as solar or wind are environmentally favourable, they are often inconsistent in their outputs. Nuclear Power is capable of providing the consistent outputs required for increased global energy demands and production requirements. There are two types of nuclear energy, fission and fusion. In both types, the total rest mass of the reaction products can be lower than that of the initial nuclei. The corresponding mass defect is released as energy according to Einstein's mass-energy relation³

$$\Delta E = \Delta mc^2. \tag{1}$$

Nuclear fission has been in commercial use since the mid 1950s, and its energy production has grown since. The main issue with current nuclear reactors is that they produce long lived radioactive waste, which is one of the reasons why the introduction of fusion is beneficial, as it avoids the same long lived waste and produces no direct carbon dioxide emissions.[4] Additionally, the energy produced from a kilogram of fuel is 4 times greater than that of fission reactors, and almost 4 million times greater than that of Fossil fuels.[5]

Despite the advantages fusion offers, it remains technically challenging to achieve since the reacting plasma must be confined at extreme temperatures. This requires methods such as inertial or magnetic confinement. Among these, magnetic confinement is the most developed approach for sustained energy production[6]. In this method, ionised plasma is confined using strong magnetic fields, preventing contact with material boundaries.

Although magnetic confinement provides a promising approach to controlled fusion, it remains a challenge to accurately predict the motion of the confined plasma. Both Collisional effects and perturbations from the electromagnetic field influence the behaviour of the charged particles, thus affecting its transport and reducing the effectiveness of confinement. It is for this reason that numerical modelling is introduced, to better understand how the particles' trajectories evolve over time under different assumptions.

This report builds on the reduced tokamak mapping model developed by Souza et al. The deterministic form of the map was numerically reimplemented from the published equations and parameter values to provide a collisionless reference case. This reproduction was then extended to include the phenomenological stochastic treatment used, in which collision-like effects are represented by random perturbations with prescribed probability and amplitude. These two recreated models provide the baseline against which the new stochastic treatment developed in this work is compared.

The original scope of the project considered progressing towards a more physically parameterised collision model in which stochastic transport could depend on plasma properties such as temperature, impurity species, or collisionality. Introducing several species- and plasma-dependent effects simultaneously, however, would make it difficult to isolate the influence of the stochastic transport mechanism itself. The scope was therefore narrowed to an intermediate problem in which the phenomenological map-space perturbation is replaced by a controlled isotropic cross-field diffusion process characterised by a single dimensionless diffusion parameter, $\hat{D}$. In this project, $\hat{D}$ is intentionally kept dimensionless and is not assigned to a particular particle species or dimensional diffusion coefficient.

The aim of this work is therefore to determine how increasing dimensionless cross-field diffusion alters the deterministic escape structure of the Souza map. In particular, the study examines how the deterministic Left/Right escape basins become probabilistic as stochastic transport is increased, and whether the resulting uncertainty remains associated with the underlying deterministic basin geometry. The resulting research question is: *how does increasing dimensionless cross-field diffusion transform the deterministic Left/Right escape structure of the Souza tokamak map?*

It is hypothesised that sufficiently weak diffusion will preserve the large-scale deterministic organisation and introduce uncertainty primarily near the deterministic Left/Right basin boundary. As the diffusion strength increases, the region in which both exits are accessible from the same initial condition is expected to broaden and the deterministic boundary is expected to become progressively less predictive.

2 Magnetic Field Structure and Reduced Phase-Space Models

2.1 Magnetic confinement in a tokamak

A tokamak is a device with a toroidal shaped chamber designed to sustain the conditions required for nuclear fusion by confining high temperature plasma, preventing it from coming into contact with material walls. This confinement is achieved through a series of magnetic field components, guiding the charged particles' motion. There are two main components present: a toroidal field generated by external magnetic-field coils, and a poloidal field produced by the plasma-induced current as well as external poloidal-field coils⁷. Their combination produces helical field lines, creating a path for the charged particles in the plasma as a result of the Lorentz force⁸,

$$m \frac{d\mathbf{v}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (2)$$

This motion can be decomposed into a rapid spiral around the magnetic field line, and a slower motion of the particle's guiding centre. For the purposes of this project, the rapid spiral is not resolved explicitly. Instead, a reduced description of the guiding centre dynamics is used, which allows the longer-term phase-space and transport behaviour to be observed using a discrete mapping model.

In an ideal tokamak, magnetic field lines lie on fixed toroidal surfaces known as magnetic flux surfaces⁹. These surfaces restrict radial motion, contributing to plasma confinement. The helicity of the magnetic field lines is commonly described by the safety factor q , which represents the relative number of toroidal and poloidal turns made by a field line. Rational values of q correspond to resonant magnetic surfaces, where perturbations can produce magnetic islands. If adjacent resonances interact or overlap, the ordered magnetic surfaces can become increasingly distorted and chaotic, allowing enhanced radial transport^{10,11}. The regular and chaotic structures produced by these dynamics can be examined using Poincaré sections, which are introduced in the following subsection.

2.2 Poincaré sections and phase-space structure

The regular and chaotic structures described above can be visualised using a Poincaré section. Rather than following the complete continuous motion of a particle, its position is recorded each time the trajectory returns to a chosen cross-section of the tokamak. Repeating this over many toroidal transits produces a set of points that shows how the trajectory moves through phase space.

The appearance of these points depends on the type of motion present. Regular trajectories form smooth closed curves, indicating that the motion remains organised around intact magnetic surfaces. Resonances produce distinct island structures, while chaotic trajectories appear as scattered points covering a wider region of phase space. Poincaré sections therefore provide a way to identify the transition between regular, resonant, and chaotic motion in the reduced tokamak model^{11,12}.

Figure 1 below shows the deterministic Poincaré section generated from the numerical reimplementation of the Souza map at the perturbation amplitude used for the later stochastic calculations,

$$\phi = 8.74 \times 10^{-3}. \quad (3)$$

The section contains several clearly defined island structures surrounded by regions of scattered trajectories. At larger values of I , particularly towards the plasma edge at $I = 1$, the regular structures become less prominent and the motion is increasingly dominated by scattered points.

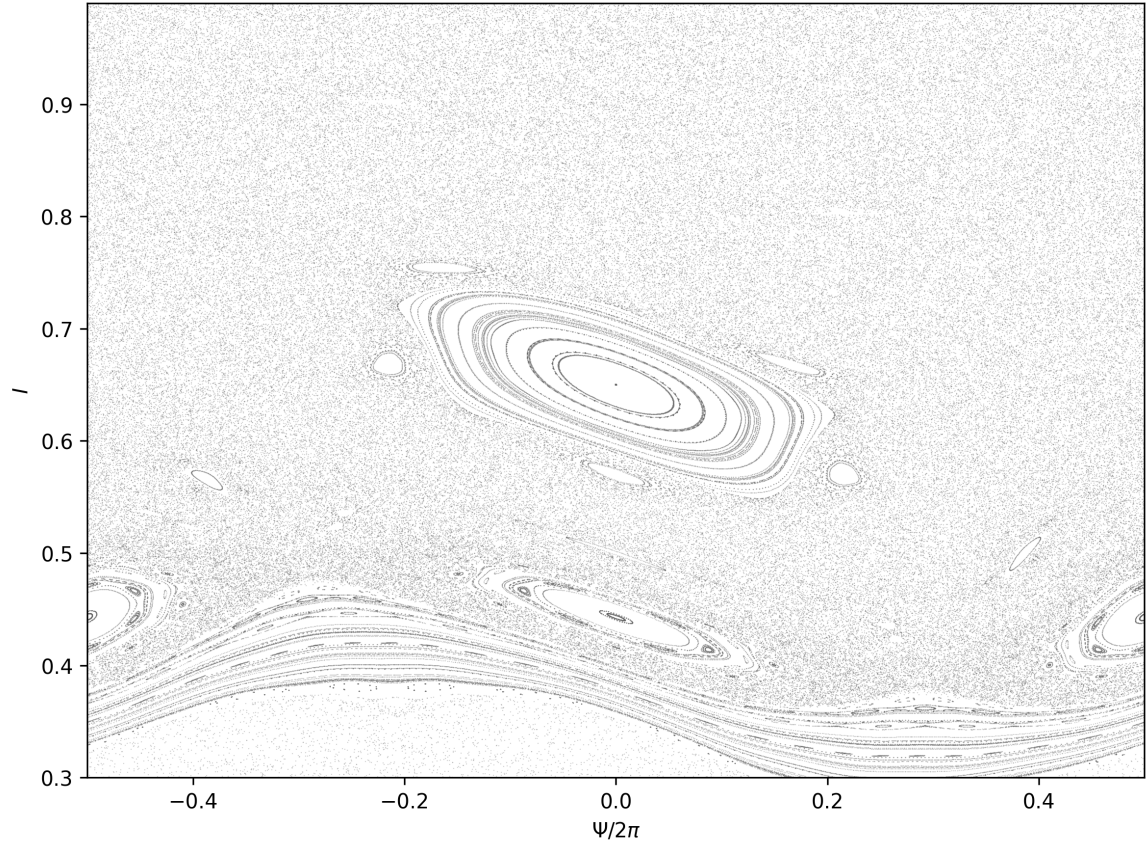


Figure 1: Deterministic Poincaré section generated from the reimplemented Souza map for $\phi = 8.74 \times 10^{-3}$. Smooth closed curves and island structures correspond to regular or resonant motion, while scattered points indicate chaotic trajectories. The coexistence of these structures provides the deterministic phase-space reference for the later stochastic calculations.

The later stochastic escape calculations focus on the edge region

$$0.80 \leq I \leq 0.99, \quad (4)$$

where the deterministic phase-space structure is already largely chaotic.

This mixture of regular and chaotic behaviour is important when considering particle transport. Regular structures can restrict radial motion, whereas chaotic trajectories are able to explore a broader region of phase space. Establishing this deterministic structure first therefore provides a reference for examining how the later stochastic models modify particle transport and escape behaviour. The mathematical mapping used to generate these dynamics is introduced in the following section.

3 The Souza Tokamak Escape Map

3.1 Normalised coordinates and deterministic map

The deterministic reference model follows the reduced mapping used by Souza et al.¹³. The radial action coordinate is

$$I = \left(\frac{r}{a} \right)^2, \quad (5)$$

where r is the minor radial position and a is the minor radius scale used to normalise the radial coordinate. The angular variable is

$$\Psi = M\theta - L\varphi, \quad (6)$$

where θ and φ are poloidal and toroidal angles and M and L are integer mode numbers. The symbol ϕ used below denotes the electrostatic perturbation amplitude and is distinct from the toroidal angle φ .

The map is written

$$I_{n+1} = I_n + K \sin \Psi_n, \quad (7)$$

$$\Psi_{n+1} = \Psi_n + F(I_{n+1}) \pmod{2\pi}, \quad (8)$$

where

$$K = \frac{4\pi M \phi}{\omega}. \quad (9)$$

To iterate the map, radial profiles are required for the safety factor, radial electric field, and parallel plasma velocity. These profiles were adopted from the model used by Souza et al., where they were chosen to represent the operating conditions of the TCABR tokamak¹³.

The safety-factor profile is

$$q(I) = 5.0 - 6.3I^2 + 6.3I^3, \quad (10)$$

the normalised radial electric-field profile is

$$\bar{E}_r(I) = 10.7I - 15.8\sqrt{I} + 4.13, \quad (11)$$

finally, the normalised parallel-velocity profile is

$$v_{\parallel}(I) = -9.867 + 17.47 \tanh(10.1I - 9.00). \quad (12)$$

The deterministic phase drift is

$$F(I) = \frac{2\pi a_0 v_{\parallel}(I)}{\omega R_0} \frac{M - Lq(I)}{q(I)} - \frac{2\pi M}{\omega} \frac{\overline{E}_r(I)}{\sqrt{I}}. \quad (13)$$

The parameters used in the present calculation are

$$M = 15, \quad L = 6, \quad \omega = 16.36, \quad a_0 = 0.18, \quad R_0 = 0.615, \quad (14)$$

with primary perturbation amplitude

$$\phi = 8.74 \times 10^{-3}. \quad (15)$$

The deterministic map is retained without modification as the reference system. This is necessary as the later stochastic calculation is intended to isolate the effect of diffusion, rather than simultaneously changing the other deterministic dynamics.

3.2 Escape definition

The normalisation in Eq. (5) places the radial escape boundary at

$$I > 1. \quad (16)$$

After a complete map period, the phase is wrapped to

$$\Psi_w \in (-\pi, \pi]. \quad (17)$$

The two exits are then defined by

$$L : \quad \Psi_w < 0, \quad (18)$$

and

$$R : \quad \Psi_w \geq 0. \quad (19)$$

The escape test is performed only after the complete map period. This event convention is retained in the stochastic extension so that differences between the models arise from stochastic transport rather than from a change in the definition or timing of escape.

3.3 Deterministic escape basin

For each initial condition, the deterministic map is iterated until the trajectory either escapes or reaches the maximum iteration count. The initial condition can then be classified according to whether the particle leaves through the Left or Right exit. Plotting these classifications across the initial condition domain produces what's called an escape basin. Such escape basins are useful in these types of nonlinear systems, as fractal basin boundaries quantify sensitivity of the final state, possibly demonstrating that two initially close trajectories may ultimately leave through different exits¹⁴.

Figure 2 shows the deterministic Left/Right escape basin in the edge-region domain used for the final calculations. The deterministic basin boundary B_0 is shown in black.

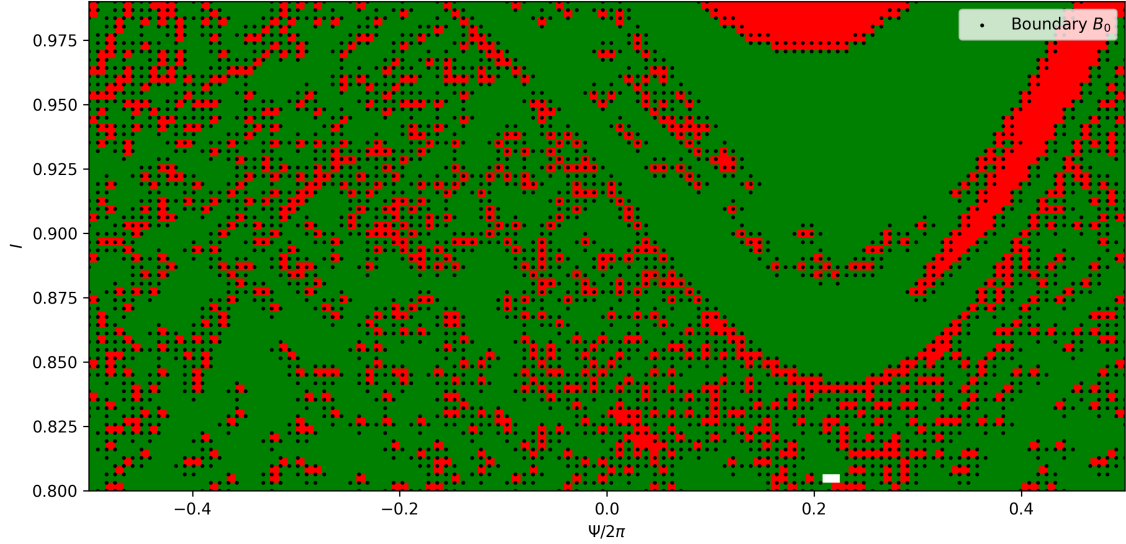


Figure 2: Deterministic Left/Right escape basin for $\phi = 8.74 \times 10^{-3}$ over the edge-region domain $0.80 \leq I \leq 0.99$. Green and red denote initial conditions that escape through the Left and Right exits, respectively. Black points represent the discrete deterministic basin boundary B_0 .

At the final 60×120 grid resolution, 5900 initial conditions escape through the Left exit, 1298 through the Right exit, and only 2 survive to the maximum iteration count. The boundary is highly distributed through the edge-region domain, demonstrating the sensitivity of deterministic escape to initial conditions.

3.4 Existing phenomenological stochastic treatment

The stochastic model considered by Souza et al. represents collision like effects using random perturbations with prescribed probability P and displacement amplitude ρ ¹³. This approach provides a phenomenological method for investigating the blurring of deterministic structures, but ρ specifies a specific displacement applied directly to the variable I and Ψ , rather than a physical cross-field diffusion coefficient.

Figure 3 compares the deterministic reference with the existing stochastic control used in the present code. The control uses

$$P = 1, \quad \rho = 10^{-3}. \quad (20)$$

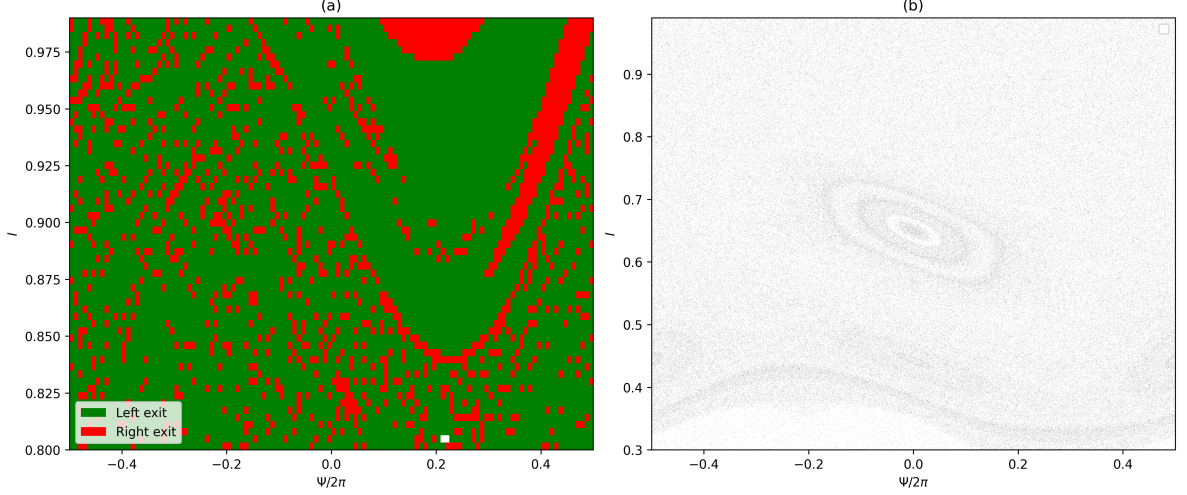


Figure 3: Comparison between the deterministic reference and the existing phenomenological Souza noise model. (a) Deterministic Left/Right escape basin for $\phi = 8.74 \times 10^{-3}$. (b) Phase-space trajectories obtained using the Souza stochastic perturbation with $P = 1$ and $\rho = 10^{-3}$. The parameter ρ belongs to the original stochastic model and is not identified with the dimensionless diffusion strength $\hat{D}$ introduced in this work.

The Souza stochastic model is retained only as a reference treatment. The parameter ρ should not be interpreted as equivalent to the dimensionless diffusion strength introduced in the following section.

4 Dimensionless Cross-Field Diffusion Model

4.1 Diffusion strength

A non-negative dimensionless diffusion parameter is introduced,

$$\hat{D} \geq 0. \quad (21)$$

For a physical cross-field diffusion coefficient $D_{\perp}$, a possible dimensionalisation would take the form

$$\hat{D} = \frac{D_{\perp} \tau_{\text{map}}}{a^2}, \quad (22)$$

where τ_{map} is the physical time represented by one complete map iteration and a is the radial normalisation scale. However, no value of $D_{\perp}$, τ_{map} , or particle species is assigned in this project. For the purposes of this study, $\hat{D}$ is treated purely as a dimensionless control parameter.

A diffusion description is a deliberate reduction of the complete plasma transport problem. Actual cross-field transport in tokamak edge plasmas can include turbulence, intermittent events, and non-diffusive behaviour¹⁵. The diffusion model is therefore intended as a simplified description of cross-field stochastic motion rather than a complete model of plasma transport. Using a single dimensionless parameter allows the effect of increasing diffusion to be studied without simultaneously introducing additional species- or plasma-dependent variables.

4.2 Cartesian Brownian diffusion substep

The normalised physical radius is

$$R = \frac{r}{a} = \sqrt{I}. \quad (23)$$

One complete inter-kick interval is divided into N_c substeps,

$$\Delta u = \frac{1}{N_c}. \quad (24)$$

For each substep, two independent standard normal random variables are generated,

$$\eta_x, \eta_y \sim \mathcal{N}(0, 1). \quad (25)$$

The Cartesian diffusion increment is

$$X' = R + \sqrt{2\hat{D}\Delta u} \eta_x, \quad (26)$$

$$Y' = \sqrt{2\hat{D}\Delta u} \eta_y. \quad (27)$$

The new action coordinate is

$$I' = X'^2 + Y'^2, \quad (28)$$

while the stochastic angular displacement is

$$\Delta\theta = \text{atan2}(Y', X'). \quad (29)$$

The phase is updated according to

$$\Psi' = \Psi + M\Delta\theta. \quad (30)$$

These increments correspond to isotropic two-dimensional Brownian diffusion with constant diffusion strength over the dimensionless substep¹⁶. Performing the update in Cartesian radial coordinates also guarantees $I' \geq 0$ as I' is reconstructed as a squared radial distance.

4.3 Strang splitting

Between the radial kicks of the Souza map, the deterministic phase evolves according to

$$\frac{d\Psi}{du} = F(I), \quad \frac{dI}{du} = 0. \quad (31)$$

The deterministic phase evolution and stochastic diffusion are combined using symmetric Strang splitting¹⁷. The interval between two successive map kicks is divided into N_c substeps, indexed by $j = 0, \dots, N_c - 1$.

For each substep, the phase is first advanced through half of the deterministic drift,

$$\Psi_{j+\frac{1}{2}} = \Psi_j + \frac{1}{2}F(I_j)\Delta u. \quad (32)$$

The Cartesian diffusion update from Eqs. (27)–(31) is then applied. This changes the radial coordinate from I_j to I_{j+1} and changes the phase by the stochastic angular displacement,

$$\Psi_{j+\frac{1}{2},D} = \Psi_{j+\frac{1}{2}} + M\Delta\theta_j. \quad (33)$$

The second half of the deterministic drift is then applied using the updated radial coordinate,

$$\Psi_{j+1} = \Psi_{j+\frac{1}{2},D} + \frac{1}{2}F(I_{j+1})\Delta u. \quad (34)$$

The two deterministic half-steps have the same form, but the first is evaluated before diffusion and the second after diffusion. This symmetric sequence is the basis of the Strang-splitting method.

A complete stochastic map period begins with the original Souza radial kick,

$$I_{n,\text{kick}} = I_n + K \sin \Psi_n. \quad (35)$$

The post-kick value $I_{n,\text{kick}}$ is then used as the initial radial coordinate for the N_c drift–diffusion substeps. After all N_c substeps have been completed, the resulting state is identified with (I_{n+1}, Ψ_{n+1}) , and the escape condition is evaluated.

A crucial implementation requirement is exact zero-noise recovery. When

$$\hat{D} = 0, \quad (36)$$

the original deterministic map is used directly rather than applying the drift diffusion routine. This ensures that the zero-diffusion case provides an exact numerical reference for the original Souza model. The agreement between the two implementations is examined later as part of the numerical verification.

4.4 Model-validity limit

The reduced Souza model is being applied to the edge-region dynamics considered in this work. For the stochastic calculation, a trajectory is therefore recorded as a model-validity failure if

$$I < 0.2. \quad (37)$$

If this limit is reached, the calculation for that trajectory is stopped and the outcome is recorded separately as a validity failure, denoted by V . This allows these trajectories to remain visible in the final statistics without assigning them an artificial Left or Right escape outcome.

4.5 Finite-time stochastic outcomes

For each initial condition

$$x = (I, \Psi), \quad (38)$$

the stochastic trajectory is repeated R_{MC} times.

Four finite-time outcomes are recorded:

$$L, \quad R, \quad S, \quad V, \quad (39)$$

representing Left escape, Right escape, survival to the maximum iteration count, and model-validity failure, respectively.

Their local probabilities are

$$q_k(x) = \frac{N_k(x)}{R_{\text{MC}}}, \quad k \in \{L, R, S, V\}, \quad (40)$$

and should satisfy

$$q_L + q_R + q_S + q_V = 1. \quad (41)$$

When at least one trajectory has escaped through a physical exit, the probabilities conditioned on physical escape are

$$\tilde{q}_L = \frac{q_L}{q_L + q_R}, \quad (42)$$

and

$$\tilde{q}_R = \frac{q_R}{q_L + q_R}. \quad (43)$$

These quantities are finite-time committor-like probabilities: they estimate the probability that an ensemble started from the same initial condition reaches one physical exit rather than the other. Committor probabilities are widely used to characterise competing outcomes in stochastic dynamical systems [18].

4.6 Stochastic exit entropy

The local stochastic exit entropy is defined by

$$h_{\text{exit}} = -\tilde{q}_L \ln \tilde{q}_L - \tilde{q}_R \ln \tilde{q}_R. \quad (44)$$

If one exit occurs with probability one, then

$$h_{\text{exit}} = 0. \quad (45)$$

If both exits are equally probable,

$$\tilde{q}_L = \tilde{q}_R = \frac{1}{2}, \quad (46)$$

then the entropy reaches its maximum,

$$h_{\text{exit}} = \ln 2. \quad (47)$$

The entropy therefore provides a local measure of how unpredictable the Left/Right physical exit has become.

4.7 Deterministic boundary and stochastic ambiguity width

The deterministic Left/Right boundary B_0 is extracted from the collisionless basin using a four-neighbour grid convention. The phase coordinate is periodic.

Distances are evaluated in the coordinates

$$u = I, \quad v = \frac{\Psi}{2\pi}. \quad (48)$$

The periodic phase distance is

$$d_{\text{per}}(v, v_b) = \min_{m \in \mathbb{Z}} |v - v_b - m|, \quad (49)$$

and the distance from an initial condition x to the nearest deterministic boundary point is

$$d^*(x, B_0) = \min_{b \in B_0} \sqrt{(I - I_b)^2 + d_{\text{per}}(v, v_b)^2}. \quad (50)$$

The primary ambiguity threshold is chosen as

$$p = 0.10. \quad (51)$$

An initial condition is classified as stochastically ambiguous when both conditioned physical exits have probability at least 0.10,

$$\tilde{q}_L \geq 0.10, \quad \tilde{q}_R \geq 0.10, \quad (52)$$

and when at least 90% of trajectories reach one of the physical exits,

$$q_L + q_R \geq 0.90. \quad (53)$$

The ambiguity set is therefore

$$A_{0.10}(\hat{D}) = \{x : \tilde{q}_L \geq 0.10, \tilde{q}_R \geq 0.10, q_L + q_R \geq 0.90\}. \quad (54)$$

The ambiguity-layer width is defined as the 90th percentile of the distance from ambiguous initial conditions to the deterministic boundary,

$$w_{0.10,0.90}(\hat{D}) = \text{percentile}_{90} \left[d^*(x, B_0) : x \in A_{0.10}(\hat{D}) \right]. \quad (55)$$

Unlike the absolute position of ambiguous points, this quantity explicitly measures how far stochastic uncertainty extends away from the deterministic basin interface.

5 Numerical Implementation and Verification

5.1 Final numerical parameters

The final stochastic maps use the edge-region initial-condition domain

$$0.80 \leq I \leq 0.99, \quad -0.5 \leq \frac{\Psi}{2\pi} \leq 0.5. \quad (56)$$

The final numerical settings are summarised in Table 1.

Table 1: Numerical parameters used for the final calculations.

Parameter	Value
Primary perturbation amplitude ϕ	8.74×10^{-3}
Initial-condition grid	60×120
I domain	0.80–0.99
$\Psi/(2\pi)$ domain	−0.5–0.5
Monte Carlo repeats per grid point R_{MC}	64
Diffusion substeps N_c	32
Maximum iterations n_{max}	20000
Global-scan initial conditions	1000
Global repeats per initial condition	64
Validity limit	$I < 0.2$

The dimensionless diffusion scan is

$$\hat{D} \in \left\{ 0, 10^{-6}, 3 \times 10^{-6}, 10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}, 10^{-3} \right\}. \quad (57)$$

Fixed random-number seeds are used to make the Monte Carlo calculations reproducible.

5.2 Verification strategy

The diffusion extension was tested before interpreting the final parameter scan. The following verifications were performed:

1. exact deterministic recovery for $\hat{D} = 0$,
2. isolated free-diffusion behaviour,
3. sensitivity to the number of diffusion substeps N_c ,
4. random-number reproducibility,
5. exact probability bookkeeping.

The main results are summarised in Table 2.

Table 2: Summary of numerical verification tests.

Test	Result
Zero-diffusion recovery	100 random initial states tested: zero one-step state mismatches, zero escape-label mismatches, and zero escape-iteration mismatches.
Free diffusion	Expected $\langle I(u) - I(0) \rangle = 4.0000 \times 10^{-4}$; measured 4.04444×10^{-4} , yielding 1.11% relative error.
N_c convergence	$N_c = 8, 16, 32, 64$ compared at $\hat{D} = 10^{-4}$. No systematic drift was observed between the finest resolutions; $N_c = 32$ was retained.
Random-number reproducibility	Identical seed produced exactly identical outcome and escape-iteration arrays, a different seed changed individual trajectories while preserving similar aggregate probabilities.
Probability bookkeeping	Maximum error in $q_L + q_R + q_S + q_V = 1$ was 0.0 over all stochastic maps.

For the free-diffusion test, a relative error of approximately 1–2% was taken as an acceptable level of agreement. The measured relative error was 1.11%, satisfying this criterion. For the convergence test, the results obtained with $N_c = 32$ and $N_c = 64$ differed by approximately 1.75 and 1.73 percentage points for P_L and P_R , respectively, while P_V differed by only 0.02 percentage points. No clear systematic change was observed as N_c was increased, so $N_c = 32$ was retained for the final calculations to balance between numerical resolution and computational cost.

6 Results

6.1 Deterministic reference and phenomenological noise

Figures 1–3 provide the reference cases for the diffusion results presented below. The deterministic calculations retain the mixed regular and chaotic phase-space structure of the Souza map and produce a mixed Left/Right escape basin within the sampled edge region. The phenomenological Souza-noise calculation is retained as a qualitative stochastic comparison only, since its parameters P and ρ are not directly related to the dimensionless diffusion strength $\hat{D}$ used in the present model.

6.2 Exit-conditioned committor

Figure 4 shows how the exit-conditioned Right-exit probability changes across the edge region as the diffusion strength is increased.

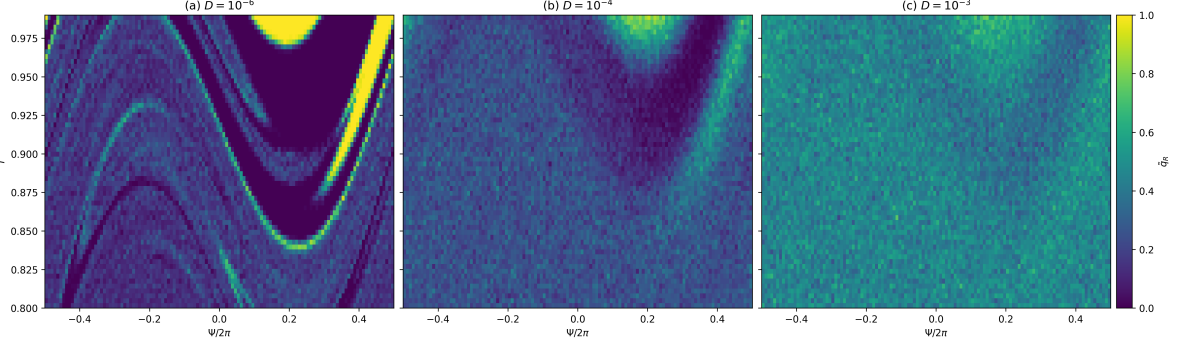


Figure 4: Exit-conditioned Right-exit probability $\tilde{q}_R$ across the sampled edge region for (a) $\hat{D} = 10^{-6}$, (b) $\hat{D} = 10^{-4}$, and (c) $\hat{D} = 10^{-3}$. Values close to 0 indicate a strong preference for the Left exit, values close to 1 indicate a strong preference for the Right exit, and intermediate values indicate increasing uncertainty between the two exits.

At $\hat{D} = 10^{-6}$, large areas remain close to either $\tilde{q}_R = 0$ or $\tilde{q}_R = 1$, and the structure resembles the organisation of the deterministic basin. This shows that weak diffusion does not immediately remove the deterministic preference for one exit. In the second panel, $\hat{D} = 10^{-4}$, the fine structure becomes visibly smoother and broader regions of intermediate probability appear. Then for $\hat{D} = 10^{-3}$, the majority of the domain contains intermediate values of $\tilde{q}_R$. This shows that repeated trajectories from the same initial condition can therefore escape through either physical exit with appreciable probability.

6.3 Stochastic exit entropy

The same transition is shown more directly by the exit entropy in Figure 5.

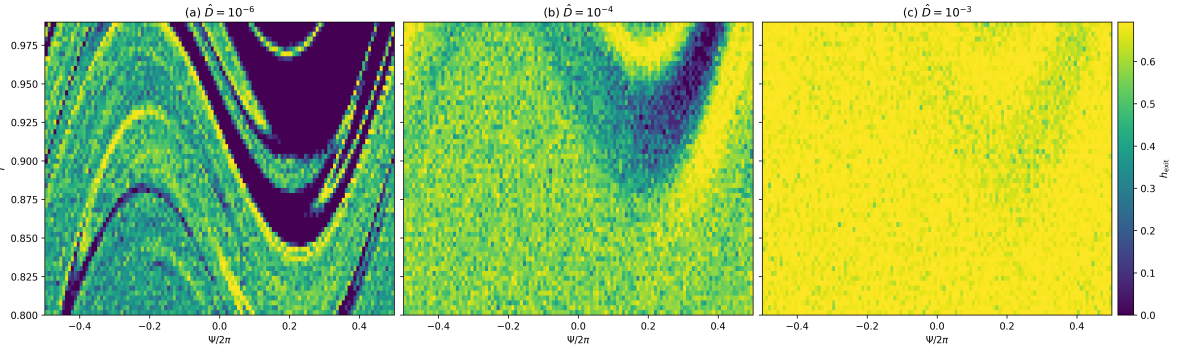


Figure 5: Exit entropy h_{exit} across the sampled edge region for (a) $\hat{D} = 10^{-6}$, (b) $\hat{D} = 10^{-4}$, and (c) $\hat{D} = 10^{-3}$. Low entropy corresponds to a preferred physical exit, while values approaching the maximum, $\ln 2$, indicate approximately equal probabilities of Left and Right escape.

At weak diffusion, high entropy regions remain structured and coexist with areas for which the exit is nearly deterministic. At $\hat{D} = 10^{-4}$, the same high entropy regions occupy a larger part of phase space. And lastly for $\hat{D} = 10^{-3}$, the entropy approaches $\ln 2$ throughout almost the full domain, indicating that the Left and Right exits have become similarly probable for many initial conditions.

The committor and entropy therefore describe the same transition from complementary perspectives. The committor identifies the preferred exit, while the entropy quantifies the uncertainty of that prediction.

6.4 Ambiguity relative to the deterministic boundary

Figure 6 shows the stochastic ambiguity set together with the deterministic boundary B_0 for $\hat{D} = 10^{-4}$. This diffusion strength was chosen as a representative intermediate case, since it lies between the weak diffusion regime and the larger diffusion values where the stochastic behaviour becomes significantly mixed.

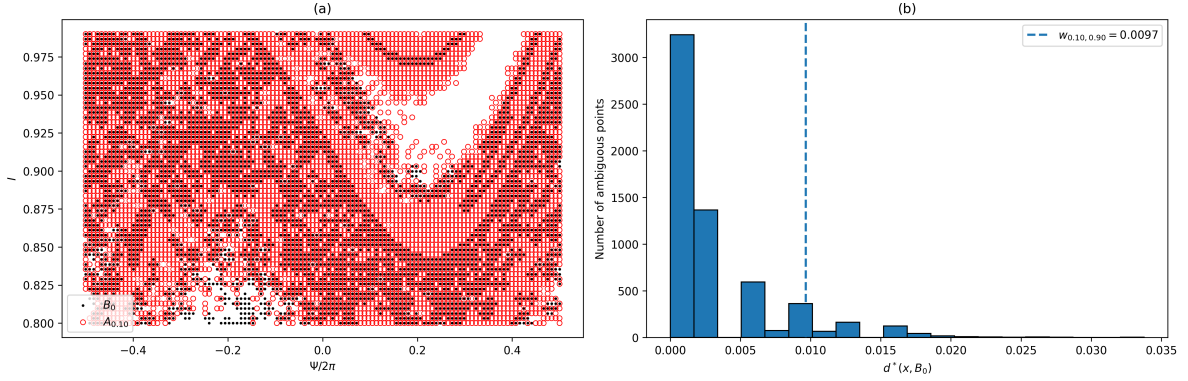


Figure 6: Ambiguity-layer analysis at $\hat{D} = 10^{-4}$. (a) Initial conditions belonging to the ambiguity set $A_{0,10}$, shown as red open circles, with the deterministic basin boundary B_0 shown in black. (b) Distribution of the periodic distance $d^*(x, B_0)$ from ambiguous initial conditions to the deterministic boundary. The dashed line marks the 90th-percentile ambiguity-layer width $w_{0,10,0.90}$.

Because the deterministic boundary itself is distributed through much of the edge domain, ambiguous points may appear spatially widespread even when they remain close to a deterministic interface. For this reason, distance from B_0 is more informative than absolute phase-space location.

At $\hat{D} = 10^{-4}$ the distance distribution is weighted towards small values and the measured 90th-percentile width is

$$w_{0,10,0.90} \simeq 9.66 \times 10^{-3}. \quad (58)$$

The ambiguity results for all non-zero diffusion strengths are shown in Table 3.

Table 3: Ambiguity-set size and 90th-percentile ambiguity width.

$\hat{D}$	Ambiguous fraction	$w_{0,10,0.90}$
10^{-6}	0.6478	6.441×10^{-3}
3×10^{-6}	0.6782	6.441×10^{-3}
10^{-5}	0.7415	6.441×10^{-3}
3×10^{-5}	0.8088	6.441×10^{-3}
10^{-4}	0.8478	9.661×10^{-3}
3×10^{-4}	0.7425	1.538×10^{-2}
10^{-3}	0.4700	2.107×10^{-2}

The ambiguity fraction initially increases as stochastic mixing becomes more widespread. It subsequently decreases at the largest values of $\hat{D}$. This decline should not be interpreted as a

recovery of deterministic behaviour. The ambiguity definition requires $q_L + q_R \geq 0.90$, while validity failures become increasingly common at high diffusion. Consequently, some stochastic initial conditions no longer satisfy the escaped-fraction requirement and are excluded from $A_{0.10}$.

6.5 Global stochastic outcomes

The integrated stochastic behaviour across the sampled edge region is shown in Figure 7.

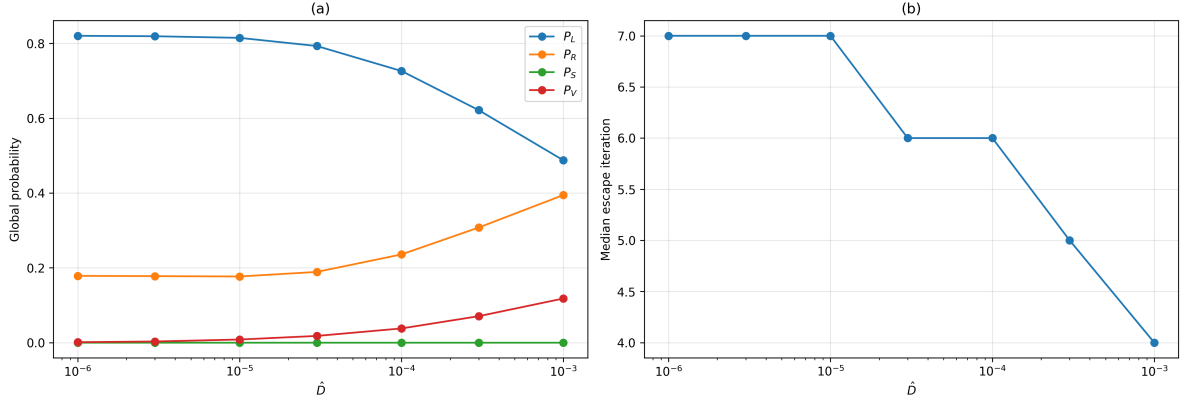


Figure 7: Global stochastic behaviour as a function of dimensionless diffusion strength $\hat{D}$. (a) Global Left-exit, Right-exit, survival, and validity-failure probabilities, P_L , P_R , P_S , and P_V . (b) Median escape iteration for trajectories that escape through a physical exit.

Figure 7 summarises the global stochastic outcomes across the sampled edge region, with the corresponding numerical values given in Table 4.

Table 4: Global finite-time stochastic outcomes.

$\hat{D}$	P_L	P_R	P_S	P_V	Median escape iteration
0	0.8120	0.1870	0.0010	0.0000	7
10^{-6}	0.8204	0.1785	0.0000	0.0011	7
3×10^{-6}	0.8192	0.1778	0.0000	0.0030	7
10^{-5}	0.8149	0.1768	0.0000	0.0083	7
3×10^{-5}	0.7931	0.1890	0.0000	0.0179	6
10^{-4}	0.7263	0.2358	0.0000	0.0379	6
3×10^{-4}	0.6215	0.3078	0.0000	0.0707	5
10^{-3}	0.4874	0.3946	0.0000	0.1180	4

The $\hat{D} = 0$ row provides the deterministic reference for the global scan. This case is not included in Figure 7 since the diffusion strength axis is logarithmic and therefore cannot display $\hat{D} = 0$.

At weak non-zero diffusion, the global probabilities remain close to the deterministic reference values. As $\hat{D}$ increases, P_L decreases while P_R increases, so the probabilities of the two physical exits become progressively more similar. At the largest diffusion strength, $P_L \approx 0.49$ and $P_R \approx 0.39$.

The global survival probability P_S is zero for all of the non-zero diffusion strengths. The small non-zero survival probability in the $\hat{D} = 0$ reference case is consistent with the deterministic escape basin, where a small number of initial conditions remain within the domain up to the

maximum iteration count. In contrast, the validity-failure probability P_V increases with diffusion strength, reaching approximately

$$P_V \approx 0.118 \quad (59)$$

at $\hat{D} = 10^{-3}$.

The median physical escape iteration also decreases from seven map periods at weak diffusion to four at the largest diffusion strength. This indicates that stronger diffusion generally causes trajectories to leave the sampled edge region more quickly.

6.6 Ambiguity-layer width

Figure 8 summarises the dependence of the ambiguity-layer width on diffusion strength.

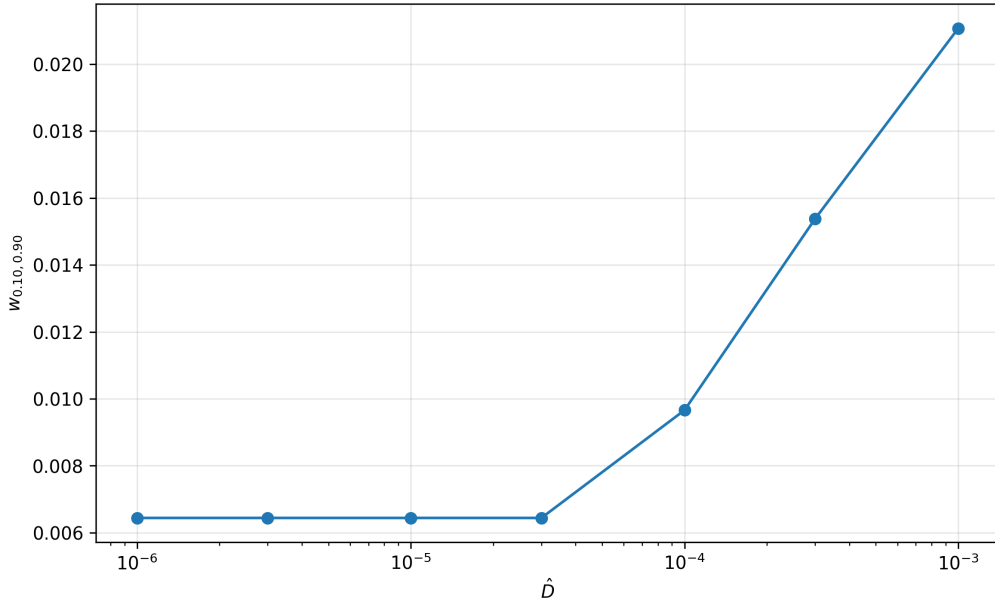


Figure 8: Ambiguity-layer width $w_{0.10,0.90}$ as a function of dimensionless diffusion strength $\hat{D}$. The width is defined as the 90th percentile of the periodic distance from ambiguous initial conditions to the deterministic basin boundary B_0 .

At the four weakest non-zero diffusion strengths, the measured ambiguity-layer width remains approximately 6.44×10^{-3} . The first resolved increase occurs at $\hat{D} = 10^{-4}$, where the width rises to 9.66×10^{-3} . It increases further to 1.54×10^{-2} at $\hat{D} = 3 \times 10^{-4}$ and reaches 2.11×10^{-2} at $\hat{D} = 10^{-3}$.

The numerical results therefore show an approximately constant width at weak diffusion followed by increasing broadening at the larger diffusion strengths. The interpretation of this behaviour, including the possible influence of grid resolution and the absence of a clear scaling law, is discussed in Section 7.

7 Discussion

7.1 Weak diffusion and deterministic escape

At weak diffusion, the stochastic model still behaves like the deterministic case. For $\hat{D} = 10^{-6}$, the committor retains spatial structure and large regions of the domain remain biased towards a single exit. The entropy map shows the same behaviour, with low entropy regions separated by smaller areas of greater uncertainty. This is consistent with the deterministic escape basin shown in Figure 2. The boundary B_0 is spread throughout the sampled edge region and separates initial conditions that lead to different final exits. In nonlinear systems with fractal basin boundaries, small perturbations can produce large variations in the final state¹⁴. In the present results, this implies that initial conditions near B_0 are the most likely to have their final exit changed by weak stochastic diffusion.

The distance-based ambiguity measure supports this interpretation. At weak diffusion, 90% of the ambiguous initial conditions are found within approximately 6.44×10^{-3} of B_0 . This shows that the original structure of the Left/Right basin still has a strong influence on the stochastic escape behaviour when the diffusion strength is small. However, there is a limitation in how the weak diffusion width should be interpreted. The same width is obtained at four consecutive diffusion strengths, while the boundary and distance calculation are based on a finite 60×120 grid. The apparent plateau may therefore be partly set by the spatial resolution of the basin calculation. It is safer to conclude that the ambiguity layer remains confined to a narrow region around the deterministic boundary than to claim that its physical width is exactly constant. A higher resolution calculation would be needed to determine whether a smaller variation exists below the current grid scale.

7.2 Crossover to broader stochastic escape

The first resolved increase in ambiguity width occurs around

$$\hat{D} \sim 10^{-4}. \quad (60)$$

This should be interpreted as a crossover within the present numerical model and parameter scan rather than as a universal value. Because $\hat{D}$ is dimensionless and has not been mapped onto a dimensional $D_\perp$, it cannot presently be associated with a particular tokamak operating condition, particle species, or measured diffusion coefficient. Despite this, the crossover appears consistently across several independent diagnostics. Near $\hat{D} = 10^{-4}$, several characteristics of the results simultaneously change. The ambiguity width increases above the weak diffusion plateau, broader regions of intermediate $\tilde{q}_R$ appear, and the entropy map becomes more uniformly high. The global Left/Right probabilities also begin to move further from their deterministic values. At the same time, the global model-validity probability is still small enough that the stochastic escape behaviour can be interpreted without being dominated by trajectories leaving the valid model region.

Taken together, these changes indicate that the behaviour around $\hat{D} = 10^{-4}$ is not being identified from a single diagnostic alone. Instead, several independent measures suggest the onset of a broader stochastic escape regime.

7.3 Loss of deterministic predictability at stronger diffusion

At $\hat{D} = 10^{-3}$, the conditioned Right-exit probability takes intermediate values over much of the domain. The corresponding entropy approaches $\ln 2$ in large regions, meaning that Left and Right escape can become almost equally probable from the same initial condition. This

represents a significant change from the deterministic model. In the collisionless basin, every initial condition has a definite final state apart from the small finite-time survivor set. Under strong diffusion, specifying the same initial condition no longer specifies a unique practical outcome, and the prediction becomes a probability distribution over possible exits. This does not make the deterministic structure irrelevant, as some spatial structure is still visible, particularly at lower and intermediate diffusion. Instead, the results show a progressive loss of predictive influence. The deterministic basin therefore becomes less useful for predicting a final state; however, its structure still influences how the stochastic escape probabilities are distributed.

7.4 Comparison with the Souza phenomenological stochastic model

The phenomenological stochastic treatment used by Souza et al. demonstrates that random collision like perturbations blur deterministic phase-space structures [13]. The present diffusion model reproduces the broader idea that stochastic motion weakens deterministic organisation, but the two models should not be interpreted as quantitatively equivalent.

The Souza control uses a perturbation probability P and fixed map-space displacement ρ . The present model instead uses Gaussian Cartesian increments whose variance is controlled by a single diffusion parameter $\hat{D}$. The displacement is therefore distributed rather than fixed, and the stochastic evolution is applied between deterministic radial kicks using multiple substeps.

The main contribution of the present model is that it provides a controlled diffusion scale together with explicitly probabilistic observables. The committor, entropy, and ambiguity width quantify how deterministic final-state sensitivity changes under increasing stochastic transport.

7.5 Validity failures and limits of interpretation

The validity outcome V becomes increasingly important at large diffusion strength. At $\hat{D} = 10^{-3}$ the global validity-failure probability is approximately 0.118, while individual initial conditions can have much larger local q_V .

This is significant as the conditioned probabilities $\tilde{q}_L$ and $\tilde{q}_R$ ignore trajectories that do not reach either physical exit. The additional requirement $q_L + q_R \geq 0.90$ was therefore imposed when defining the ambiguity set. The local survival and validity maps are provided in Appendix B. Since validity failures are still relatively uncommon around $\hat{D} = 10^{-4}$, the changes observed at this diffusion strength cannot be explained by trajectories leaving the valid model region alone.

7.6 Absence of a single scaling law

The ambiguity-width scan does not support a single power-law relationship over the full range of $\hat{D}$. Instead, the results show two broad regimes with a transition between them. At weak diffusion, the ambiguity remains confined to a narrow region close to the deterministic basin boundary and the measured width changes very little. Then at larger diffusion strengths, the ambiguity region becomes broader as stochastic effects extend further across the sampled domain. The crossover describes the transition between these two behaviours, which begins around $\hat{D} \sim 10^{-4}$ in the present results.

$$\text{weak-diffusion localisation} \longrightarrow \text{crossover} \longrightarrow \text{stochastic broadening.} \quad (61)$$

Fitting a power law only to the final few points would produce a numerical exponent, but the available range would not establish that such a scaling is necessarily correct. No power law fit is therefore reported. This restraint is significant as the weak diffusion plateau may be affected by finite grid resolution. A future calculation with a finer deterministic boundary and larger Monte Carlo ensemble would be required before attempting to infer an asymptotic scaling law.

7.7 Limitations of the present model

The model is intentionally reduced, and several limitations should be recognised.

First, $\hat{D}$ is dimensionless. Although Eq. (22) provides a possible relation to a physical $D_{\perp}$, no map time τ_{map} or species specific diffusion coefficient is assigned. The present results therefore establish only a dimensionless stochastic crossover. Second, only one primary perturbation amplitude,

$$\phi = 8.74 \times 10^{-3}, \quad (62)$$

is investigated in detail. Since the deterministic basin changes with ϕ , the location and nature of the stochastic crossover may also change. Third, the calculation uses a finite spatial grid and 64 Monte Carlo repetitions per grid point. These settings provide tractable and reproducible results, but very weak probabilities and fine boundary structure remain resolution-dependent. Finally, isotropic Brownian diffusion is only one idealised representation of stochastic cross-field motion. Experimental edge plasma transport can involve intermittent structures and non-diffusive processes [15].

7.8 Future work

The most direct extension would be to assign a physical interpretation to $\hat{D}$. This would require identifying a suitable physical map time τ_{map} and radial scale a , allowing

$$D_{\perp} = \frac{\hat{D}a^2}{\tau_{\text{map}}} \quad (63)$$

to be evaluated.

Once a physical diffusion coefficient is established, the model could be extended to investigate how $D_{\perp}$ depends on plasma conditions. This would provide a route towards the broader species- and temperature-dependent model originally motivating the project.

A second extension would repeat the diffusion scan at additional deterministic perturbation amplitudes, for example $\phi = 7.65 \times 10^{-3}$ and $\phi = 10.38 \times 10^{-3}$. This would determine whether stochastic broadening depends systematically on the deterministic phase-space structure.

Further numerical work could increase the basin resolution, increase the Monte Carlo ensemble, vary the ambiguity thresholds, and investigate whether the weak diffusion plateau persists below the current grid scale.

8 Conclusion

This project investigated how dimensionless stochastic cross-field diffusion changes the escape behaviour of the deterministic Souza tokamak map. A reduced diffusion model was introduced in which isotropic Brownian motion acts between deterministic map kicks, while the original deterministic map is recovered exactly when $\hat{D} = 0$. The stochastic implementation was verified through free-diffusion, convergence, reproducibility, and probability-bookkeeping tests.

The results show a change in escape behaviour as the diffusion strength is increased. At weak diffusion, the deterministic Left/Right basin continues to strongly influence the final outcome, with stochastic uncertainty remaining concentrated close to the deterministic boundary. Around $\hat{D} \sim 10^{-4}$, the ambiguity layer begins to broaden, while at larger diffusion strengths the Left

and Right exits become increasingly mixed and the deterministic basin becomes less useful for predicting a single final state.

The results therefore support the hypothesis that weak diffusion initially introduces uncertainty near the deterministic basin boundary before broader stochastic transport progressively reduces the predictive influence of the deterministic escape structure. The behaviour is better described as a transition from weak-diffusion localisation through a crossover to broad stochastic mixing than by a single scaling law.

Because $\hat{D}$ remains dimensionless, the value $\hat{D} \sim 10^{-4}$ should not be interpreted as a universal physical transport threshold. A future physical interpretation would require relating $\hat{D}$ to a dimensional cross-field diffusion coefficient and ultimately to specific plasma conditions. Nevertheless, the present calculation provides a controlled and quantitatively measurable intermediate step between the deterministic Souza escape map and a more complete physically parameterised stochastic transport model.

9 Appendices

A Additional Numerical Verification

A.1 N_c convergence data

Table 5 gives the complete convergence results for the number of drift–diffusion substeps used per map period.

Table 5: Convergence test at $\hat{D} = 10^{-4}$.

N_c	P_L	P_R	P_S	P_V	Median escape iteration
8	0.7175	0.2406	0.0000	0.0419	6
16	0.7188	0.2390	0.0000	0.0423	6
32	0.7298	0.2310	0.0000	0.0392	6
64	0.7123	0.2483	0.0000	0.0394	7

Between $N_c = 32$ and $N_c = 64$,

$$|\Delta P_L| = 0.0175, \quad |\Delta P_R| = 0.0173, \quad (64)$$

$$|\Delta P_S| = 0, \quad |\Delta P_V| = 2.08 \times 10^{-4}. \quad (65)$$

The Left/Right differences are of order one to two percentage points and do not show a monotonic trend with increasing N_c . The validity-failure probability is essentially unchanged between the two finest resolutions.

A.2 Random-number reproducibility

For the fixed-seed test, repeating the calculation with the same random-number seed produced identical trajectory-outcome arrays and identical escape-iteration arrays.

For one seed, the aggregate result was

$$(P_L, P_R, P_S, P_V) = (0.73271, 0.22438, 0, 0.04292), \quad (66)$$

while a different seed gave

$$(P_L, P_R, P_S, P_V) = (0.73458, 0.22688, 0, 0.03854). \quad (67)$$

The largest difference among the aggregate probabilities was

$$4.375 \times 10^{-3}. \quad (68)$$

The calculation is therefore exactly reproducible when the random seed is fixed while still producing statistically similar integrated results for a different stochastic realisation.

B Local Survival and Validity Diagnostics

The conditioned probabilities $\tilde{q}_L$ and $\tilde{q}_R$ describe only trajectories that escape through one of the two physical exits. The unconditioned survival and model-validity probabilities are therefore shown separately.

B.1 Local survival probability

Figure 9 shows the local survival probability for the same representative diffusion strengths used in the committor and entropy plots.

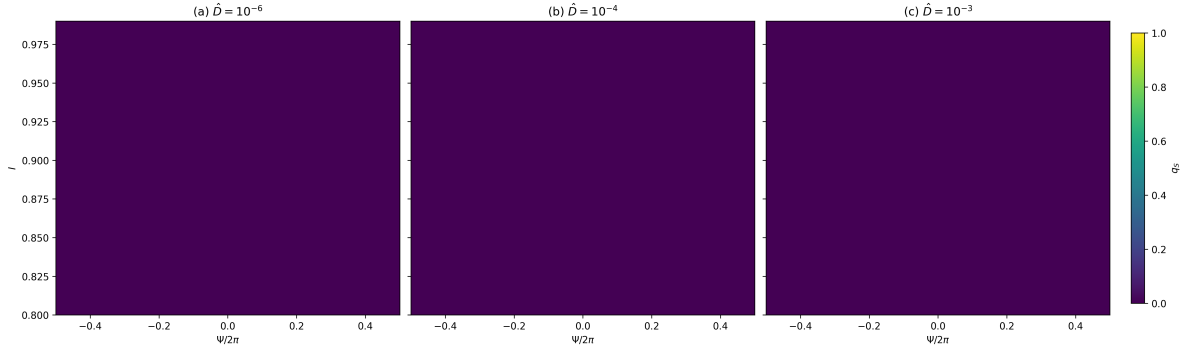


Figure 9: Local finite-time survival probability q_S for $\hat{D} = 10^{-6}$, 10^{-4} , and 10^{-3} . All three non-zero diffusion cases have $q_S = 0$ throughout the sampled edge-region domain.

For each of these non-zero diffusion values,

$$q_S = 0 \quad (69)$$

throughout the sampled domain. Finite-time survival therefore does not influence the conditioned Left/Right results at these representative diffusion strengths.

B.2 Local validity-failure probability

Figure 10 shows the local probability of reaching the model-validity boundary $I < 0.2$.

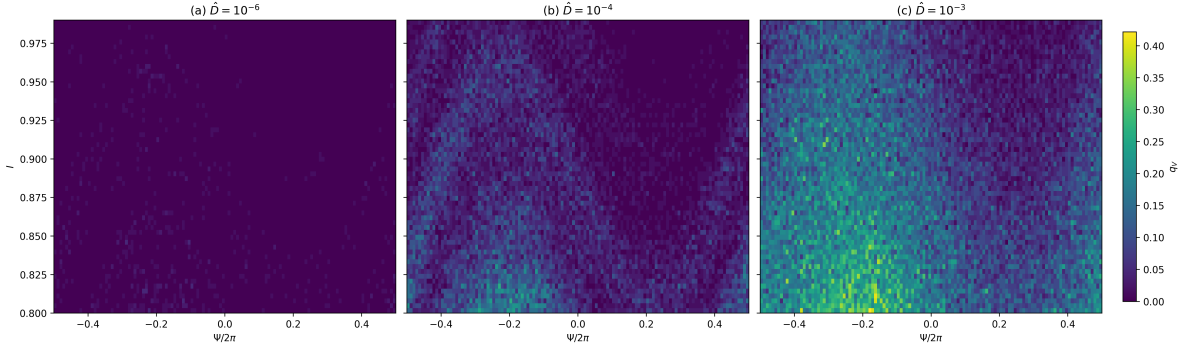


Figure 10: Local model-validity-failure probability q_V for $\hat{D} = 10^{-6}$, 10^{-4} , and 10^{-3} . The probability becomes larger and more spatially extensive as the dimensionless diffusion strength is increased.

Unlike survival, validity failure becomes increasingly significant at stronger diffusion and is spatially non-uniform. This confirms the necessity to retain V as a separate outcome and to impose the escaped-fraction condition when constructing the stochastic ambiguity set.